TIME APPLIANCES
(TAP)



Last Inch Synchronization – How to Get Time to the Software Level

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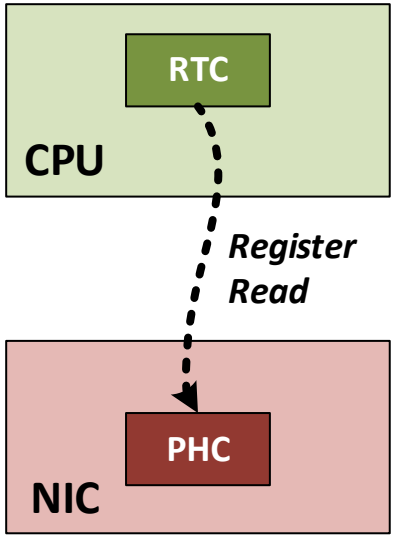
Skyworks Solutions Inc – Timings Products Business

Agenda

- Options for last-inch synchronization
- Ongoing ITU-T standardization work for data center synchronization
- Performance measurements for PTM and FLEX IO ePPS
- Factors affecting overall synchronization performance
- Call for Action

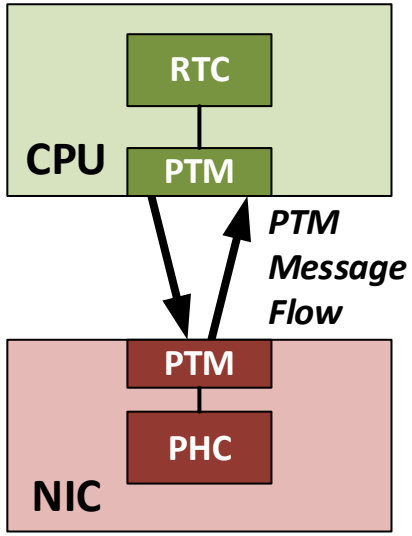
Methods for Crossing the Last Inch

Software Only



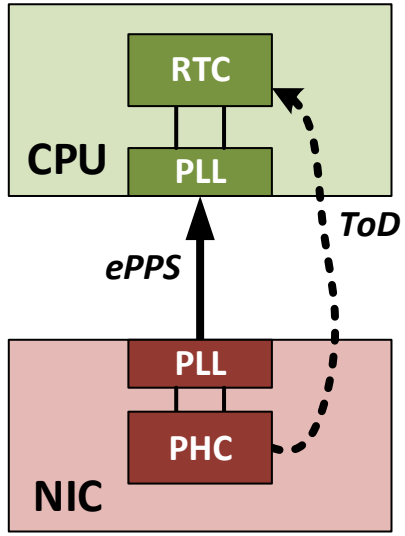
- CPU reads the time from the NIC and sets RTC
  - Really simple
  - And a really bad idea

Hardware-Assist (PTM)



- Time transport protocol similar to PTP
  - Bidirectional message flow
  - Time stamping at each end

Hardware (ePPS)



- Physical time transport
  - ePPS combines tick frequency and top-of-second
  - ToD transferred by software

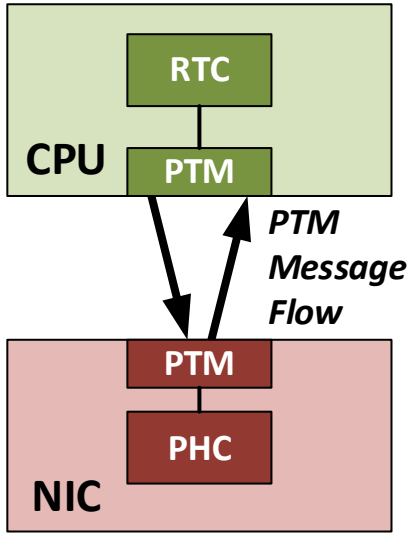
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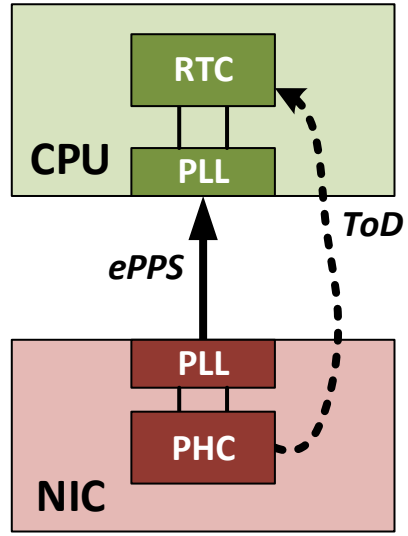
Latency too high
Jitter very CPU load dependent

Hardware-Assist (PTM)



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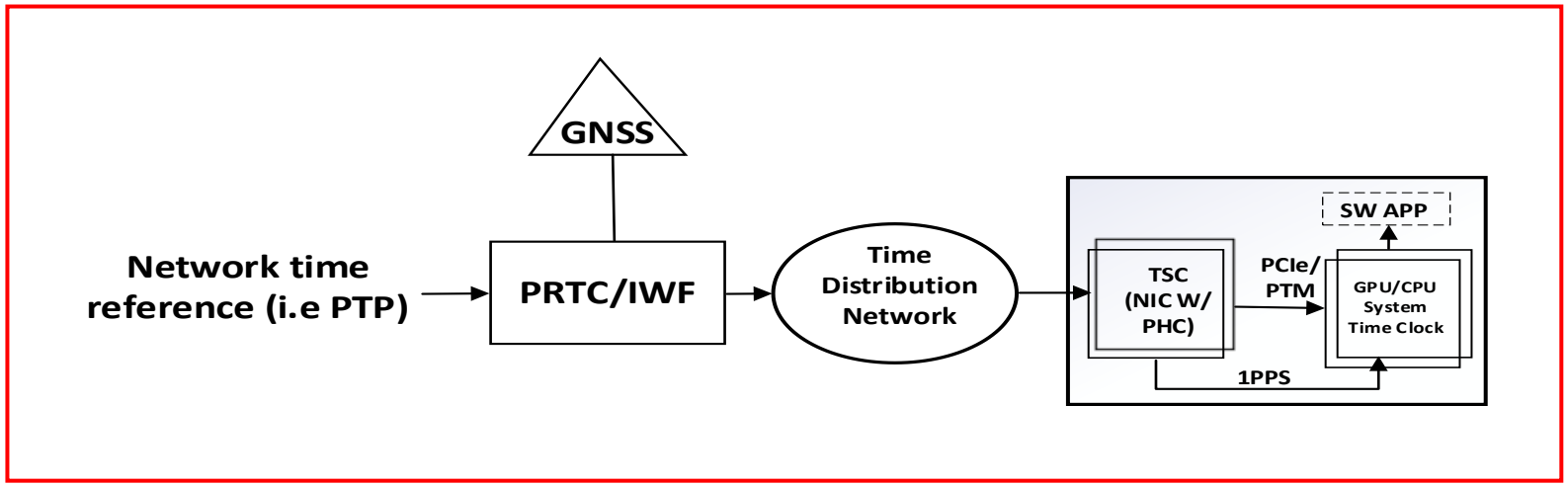
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Synchronization for Data Centers – ITU-T Activity

- ITU-T SG15/Q13 simplified HRM for data center synchronization



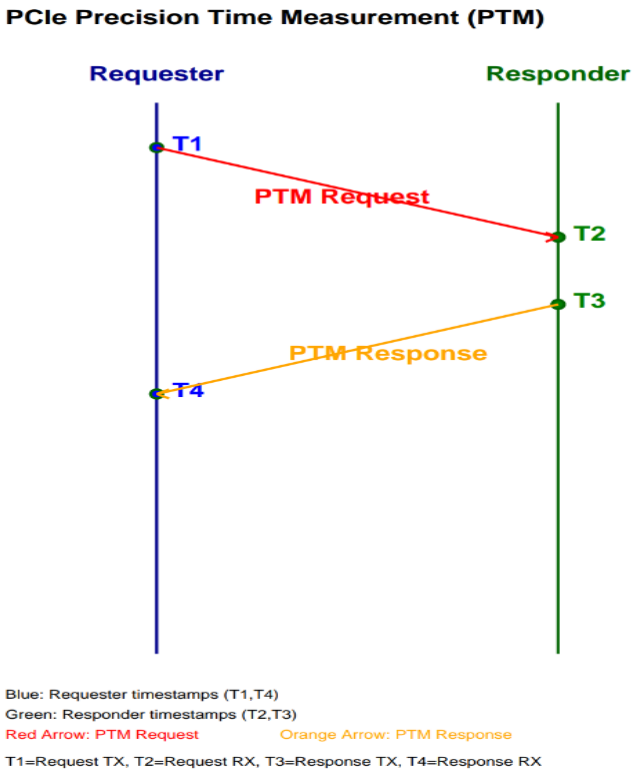
- 1PPS and PCIe PTM for last inch time distribution
  - "Last inch" referred the time distribution from NIC to GPU/CPU (the right box above)
- Time error budget for last inch time distribution
  - Accuracy class level A: +/- 2 µs, typical PCIe without PTM
  - Accuracy class Level B: +/- 200 ns, typical PCIe with PTM
  - Accuracy class Level C: +/- 50 ns, with physical edge clock signal, 1PPS/ePPS

Precise Time Measurement (PTM) Over PCIe

- PCI-SIG (PCI Express) PTM is implemented in PCIe Configuration space with a set of PTM message TLPs (Transaction Layer Packets) used to exchange timestamps
- PTM Consists of

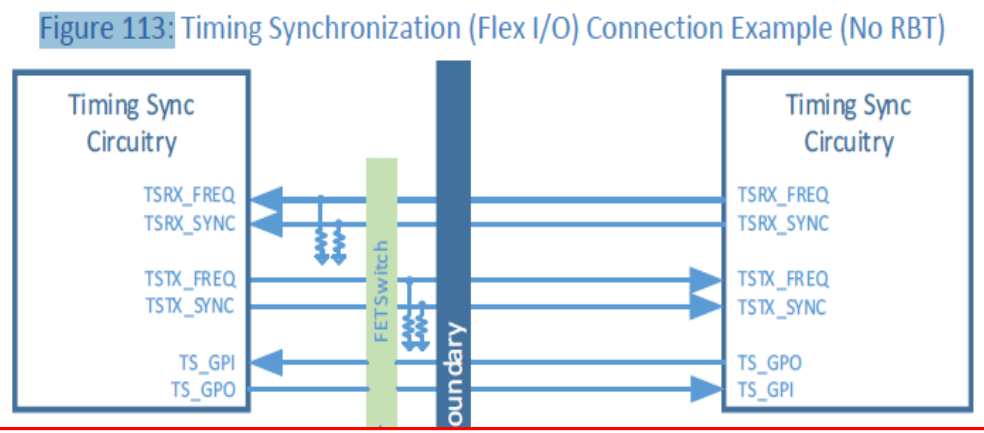
  PTM Capability Structure
  - Advertises PTM support
  - Enables/Disables PTM
  - Indicates PTM Roles (Requester/Responder)
  - Located in PCIe Extended Capabilities Space

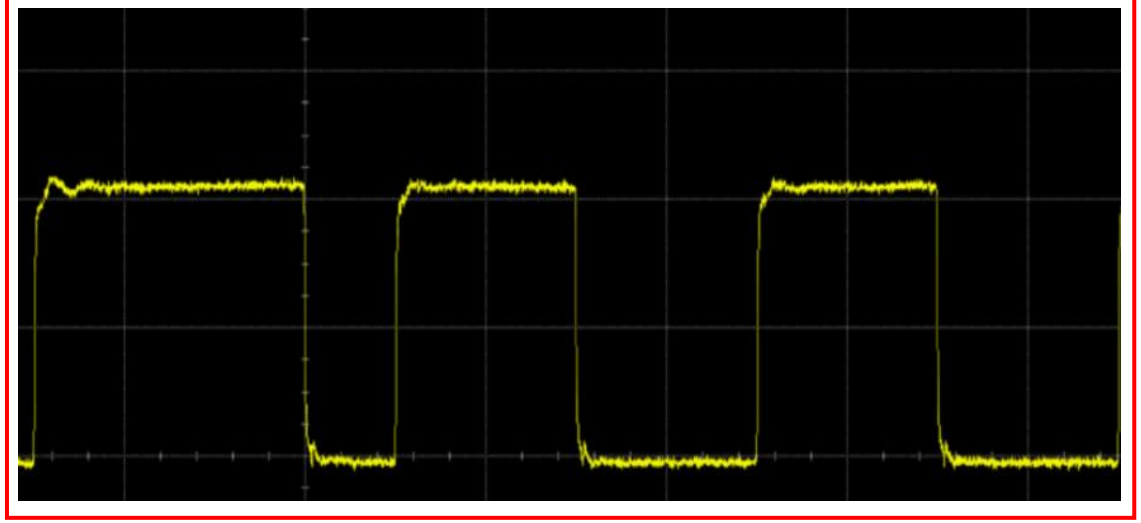
  PTM Messages
  - PTM request
  - PTM Response
  - PTM Response D(delayed response)
  - Timestamp Format as 64-bit Time Value



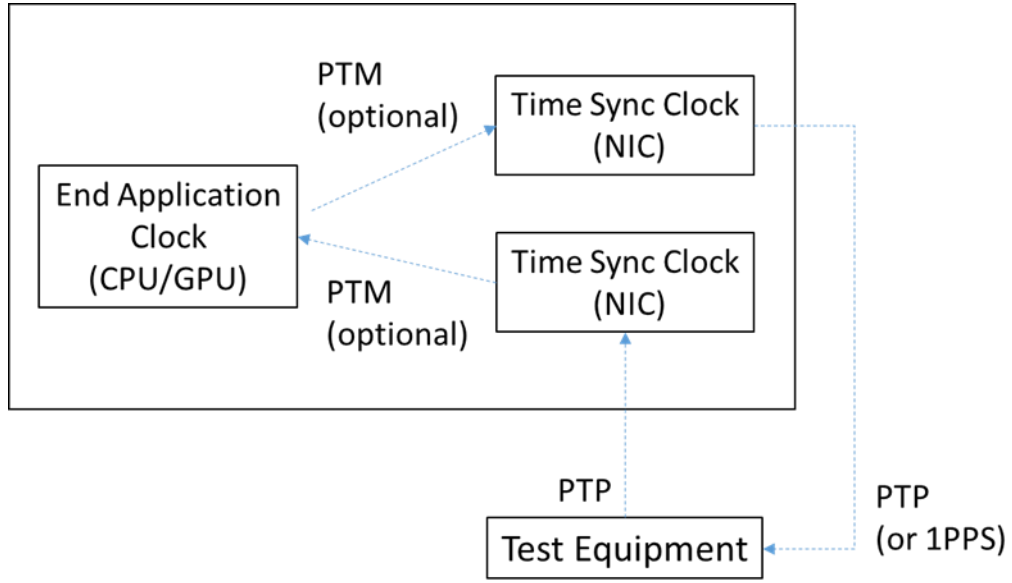
Flex IO Distribution of ePPS / 1PPS

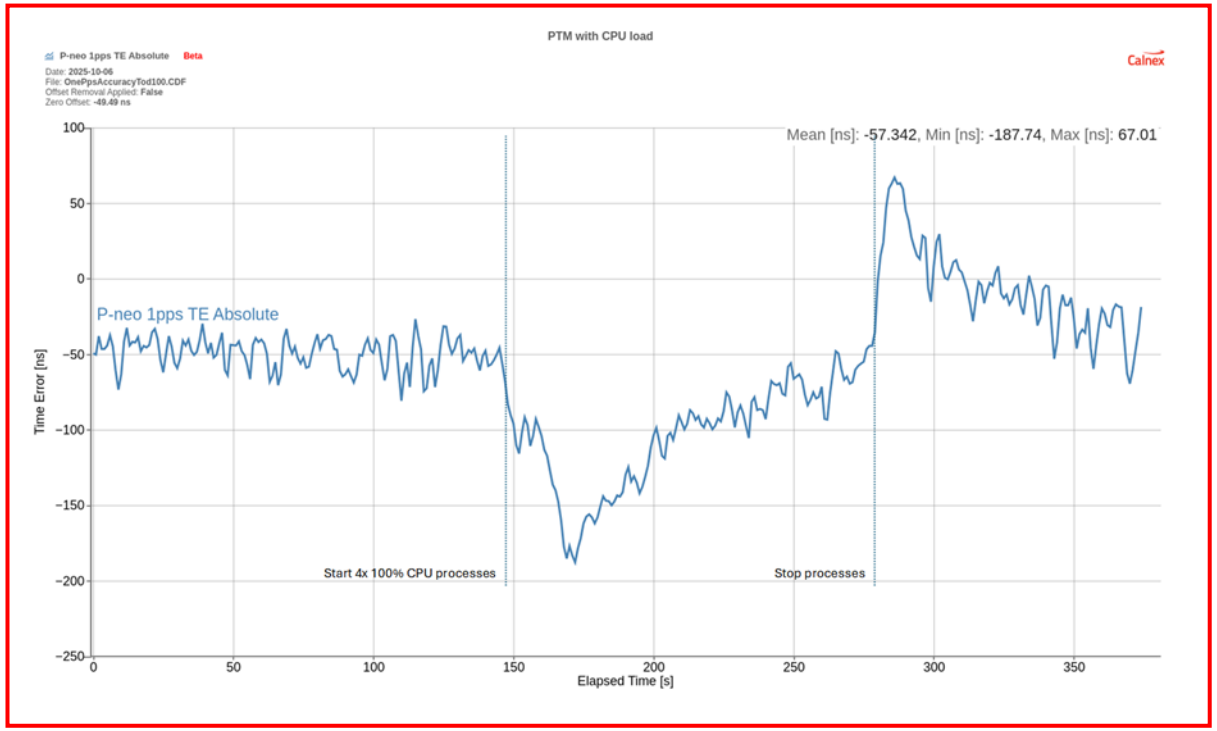
- OCP NIC specifies Flex IO to distribute the time between two timing sync circuits, such as a NIC and its associated GPU/CPU
  - TS0/1_RX_Freq: timing app received frequency clock, 10 MHz only or 10 MHz with ePPS
  - TS0/1_RX_Sync: timing app received synchronization signal, 1PPS
  - TS0/1_TX_Freq: timing app transmit frequency clock, 10 MHz only or 10 MHz with ePPS
  - TS0/1_TX_Sync: timing app transmit synchronization signal, 1PPS
- ePPS first defined by QSFP-DD MSA
  - ePPS uses varied duty cycle to indicate top-of-second
  - Underlying carrier clock can be used as “tick clock” to RTC
  - OCP NIC Flex IO defines carrier frequency of 10 MHz





PTM Performance Measurement





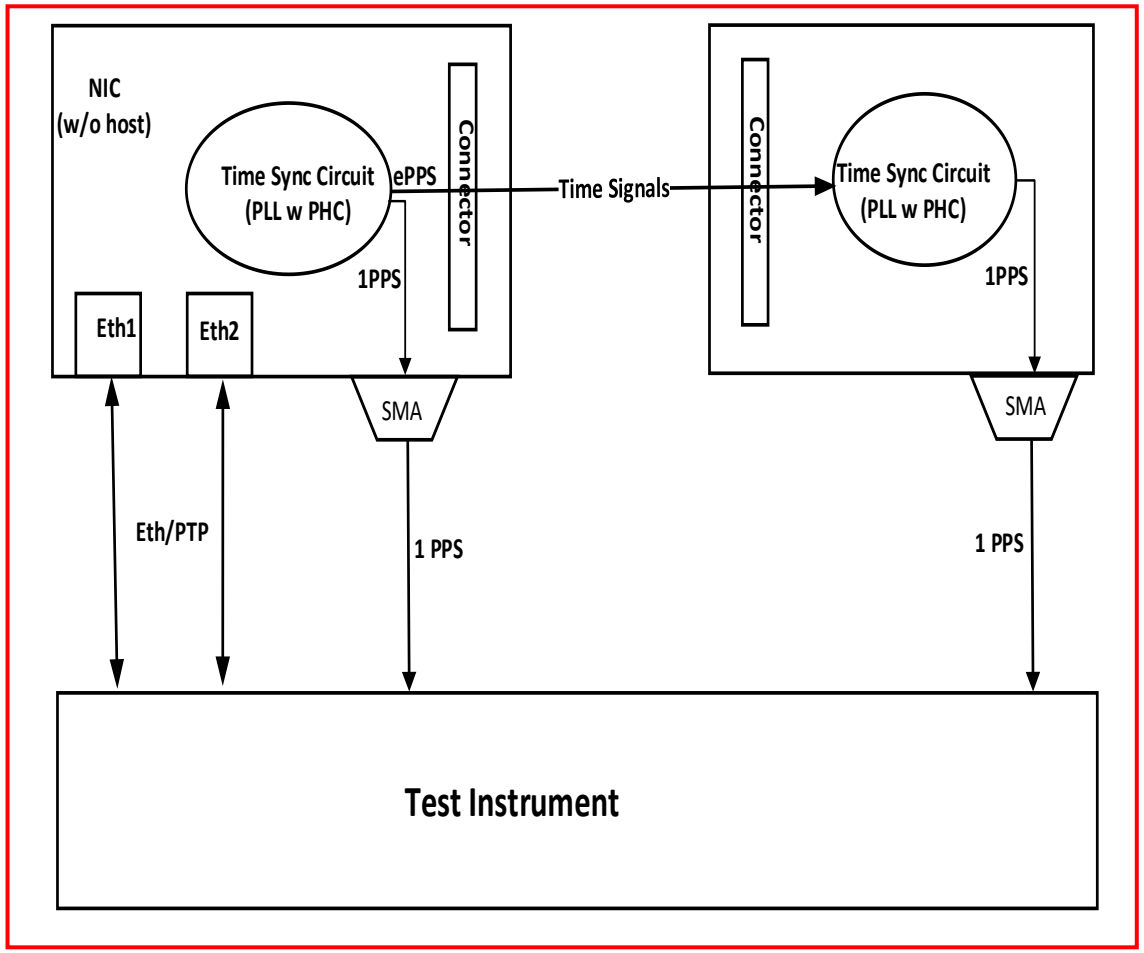
2 cascaded RTM with T-TSC in the NIC

- PTP input from test equipment to the NIC is ideal and no impairment
- The noise level significant by using PTM and it is heavily impacted by the traffic load comparing with ePPS (isolated with any traffic issue)

Source: Calnex, ITU-T SG15/Q13, WD13-20, Massy, March 2026

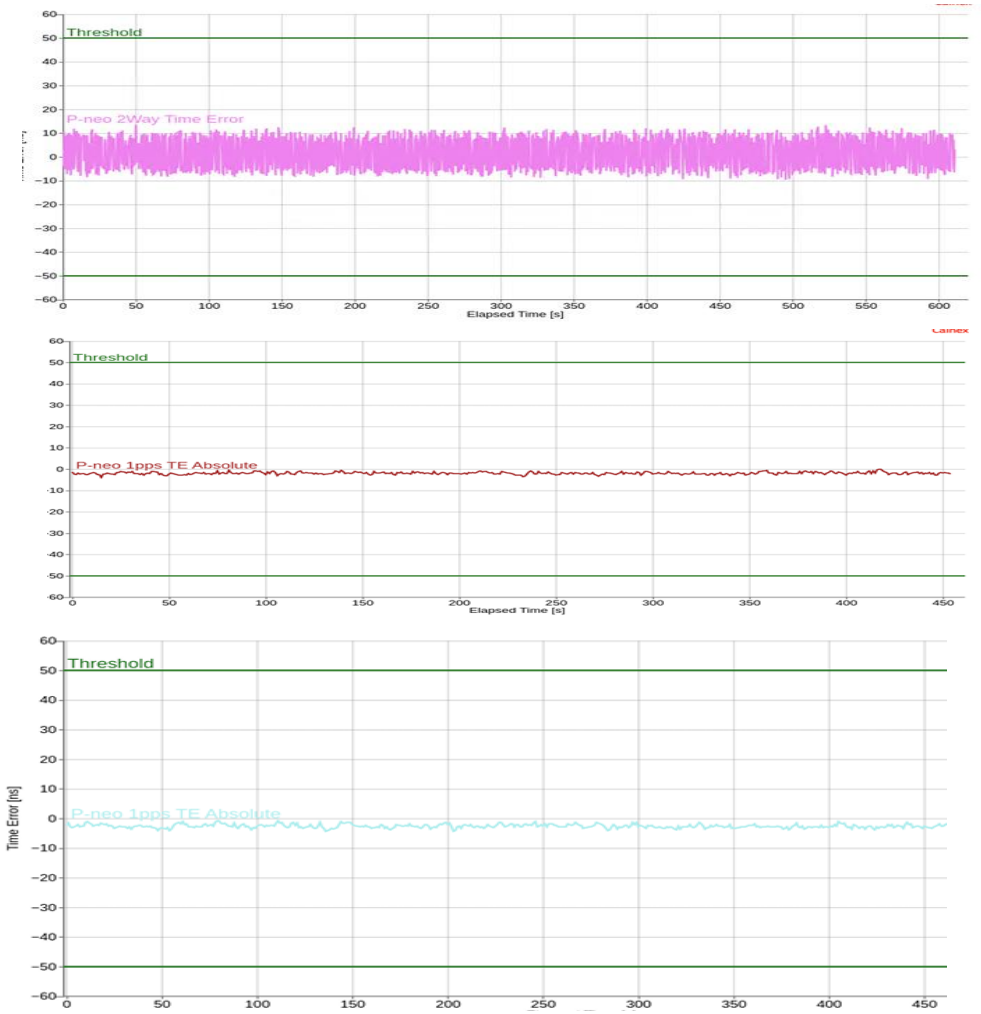
System Setup for PTP and 1PPS Measurement

- NIC can have a host attached for PTP servo
- PTP packets through ethernet interfaces, BC or TC to recovery time
  - The performance of PTP can be measured if there are two Ethernet interface via BC
  - A time sync circuit (i.e, PLL) can have PHC and therefore the frequency and 1PPS (and ToD if PHC) by software servo recovery from PTP
  - Time Sync circuit combines 10 MHz and 1PPS to generate ePPS , which is connected via I/O connectors to the second time circuit
  - The “identical” 1PPS can be generated via SMA back to the test instrument
- The second time sync circuit (i.e, PLL) receives the ePPS
  - Demodulates ePPS back to 1PPS and 10 MHz
  - 1PPS output measured against the original PTP source in the test instrument



Flex IO ePPS Performance Measurement

	Mean (ns)	Min (ns)	Max (ns)	Max-Min (p-p)
NIC PTP Output (T-BC mode)	1.47	-9.6	13.52	23.12
NIC 1PPS Output	-1.89	-3.8	-0.05	3.75
1PPS Recovered in Host From ePPS	-2.55	-4.3	-0.08	3.5
PTM Performance	Around 50 ns time error (2 PTM cascaded, using RMS, 28.8 ns to 35.ns for each PTM, 35.4 ns assuming PTP clock error ignorable)			



NOTE: There is minimal degradation between 1PPS at NIC and host
In both measurement cases (ePPS and PTM), the PTP input source is ideal without noise impairment, PTM performance is on 2 cascaded PTMs.

Key Factors Impacting Time System Performance

- Factors impacting PTP network performance
  - Majority time error from PTP network portion
  - Packet delay variation (PDV) with traffic load
  - Network asymmetry
  - Timestamp accuracy
  - Time hardware component, OCXO/TCXO
  - Packet rate and servo algorithm
  - Reduction of packet delay variation is critical

- Factors impacting “last inch” ePPS performance
  - Trace route length
  - Time circuit buffer
  - Level shifting
  - Estimation of the delay is critical for achieving adequate delay compensation

Call to Action

- Continue working with ITU-T to refine requirements
  - Can have different category of requirement (ePPS/PPS vs PTM) for the last inch synchronization?
- Implement PTM as a baseline last inch sync solution
  - In addition, standardize measurement of PTM within OCP community against a predefined limit
- Plan for PCIe Flex IO-based ePPS to support more stringent synchronization requirements
  - Such as tight scheduling of distributed processing tasks
- Work with PLL vendors their products meet the needs of ePPS
  - Consider features such as high-precision delay measurement and compensation to allow for PCB trace length

Thank You!

